EXPERIMENT NO. 7

**Implement Strings and Console Input/Output**

PART B – TO BE COMPLETED BY STUDENTS

Roll No. C-60

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Class / Division: SE-C

Batch: C2

Date of Experiment _____

Date of Submission _____

Grade _____

**B.1 Document Created by the Student**

**Repeat the experiment using Scanner and compare the input code.**

The program is modified to use the Scanner class instead of BufferedReader to accept the patient name and ID. The trim() and toUpperCase() methods process the name, while StringBuilder constructs the final report.

**B.2 Student Program / Output**

Paste/write the final program and attach or reproduce the observed output below.

```
import java.util.Scanner;
public class StringInputDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter patient name: ");
        String name = sc.nextLine();

        System.out.print("Enter patient ID: ");
```

```
        int id = sc.nextInt();

        String trimmed = name.trim();
        String upper = trimmed.toUpperCase();

        StringBuilder report = new StringBuilder();
        report.append("Patient ID: ");
        report.append(id);
        report.append(", Name: ");
        report.append(upper);

        System.out.println(report);
        System.out.println("Name length: " + trimmed.length());

        sc.close();
    }
}
```

```
Enter patient name: Mansi
Enter patient ID: 101
Patient ID: 101, Name: MANSI
Name length: 5
```

**B.3 Observations and Learning**

**● Why is String immutable?**

String is immutable because its value cannot be changed after creation. Any modification creates a new String object. This improves security, reliability, and safe sharing of String objects.

**● When would StringBuilder be preferable to String?**

StringBuilder is preferable when repeated modifications, such as appending or inserting text, are required because it modifies the same character sequence.

**● What is the difference between readLine() and token-based Scanner input?**

readLine() reads a complete line, including spaces. Token-based Scanner input using next() reads only one token up to whitespace. Scanner's nextLine() also reads a complete line.

**Student observation / learning:**

I learned to accept console input using Scanner, process strings using built-in methods, and construct a report using StringBuilder.

**B.4 Conclusion**

Input/output classes listed in the syllabus are used directly.

**Student conclusion:**

The experiment demonstrates console input using Scanner and string processing using trim(), toUpperCase(), and StringBuilder. The program successfully displays the patient ID, name, and name length.

**B.5 Questions of Curiosity**

**5. Why is String immutable?**

String is immutable because its contents cannot be modified after creation. This improves security, thread safety, and reliability.

**6. When is StringBuilder useful?**

StringBuilder is useful when strings need frequent modification, such as appending, inserting, or deleting characters.

**7. What is the difference between Scanner and BufferedReader?**

Scanner provides convenient methods to read different data types and tokens. BufferedReader reads text efficiently, usually line by line, and numeric input must be converted separately.